Project Design Phase-II
Technology Stack (Architecture & Stack)

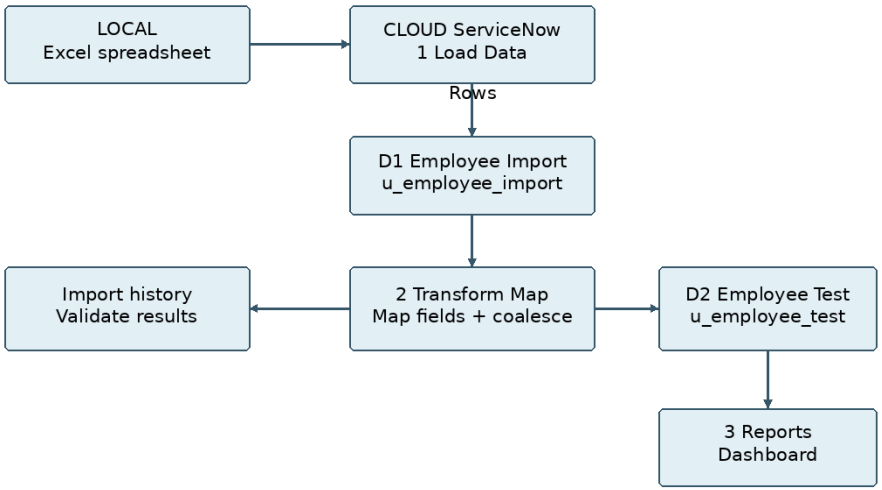
Date	1 October 2026
Team ID	SWTID-2026-9374
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows employee spreadsheet loading, field transformation, storage, validation, reporting and dashboard access.

Project: Import Data using Transform Maps (Spreadsheet)

Reference: Import Data using Transform Maps(1).docx



Architecture notes:

Load employee spreadsheets into Employee Import.
ServiceNow hosts the import and target tables.
No external API is required.
Employee Test stores validated employee records.
Reports and dashboards display employee data.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Forms to load data, configure mappings and view employee records.	ServiceNow web interface
2.	Application Logic-1	Load employee Excel rows into a staging table.	Load Data and Import Sets
3.	Application Logic-2	Map source fields and create or update target records.	Transform Maps and Mapping Assist
4.	Application Logic-3	Match Employee ID to prevent duplicate records.	Coalesce on Employee ID
5.	Database	Store Employee ID, Name, Email, Department and Location.	Employee Test (u_employee_test)
6.	Cloud Database	Stage incoming employee rows before transformation.	Employee Import (u_employee_import)
7.	File Storage	Prepare and upload Sample Spreadsheet.xlsx.	Google Sheets export; Excel file
8.	External API-1	No external API is used in the supplied project.	Not applicable
9.	External API-2	No second external API is used.	Not applicable
10.	Machine Learning Model	No machine learning model is used.	Not applicable
11.	Infrastructure (Server / Cloud)	Local browser and spreadsheet file; ServiceNow instance hosts tables, imports, reports and dashboards.	ServiceNow cloud platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is specified.	ServiceNow built-in features
2.	Security Implementations	Restrict users, groups and roles for reports and dashboards; apply authorised table access.	ACLs and dashboard sharing
3.	Scalable Architecture	Reuse the staging table and Transform Map for additional spreadsheet imports.	Import Sets and Transform Maps
4.	Availability	Cloud instance access is required; no separate availability configuration is described.	ServiceNow instance
5.	Performance	Bulk load rows, reuse mappings and prevent duplicate inserts. No benchmark is specified.	Load Data and coalesce

References:

Source document: Phases 1 to 3 - Spreadsheet, Employee Test and Import Set.

Source document: Phases 4 to 5 - Field mapping, transformation and validation.

Source document: Phases 6 to 7 - Employee ID coalesce and repeat imports.

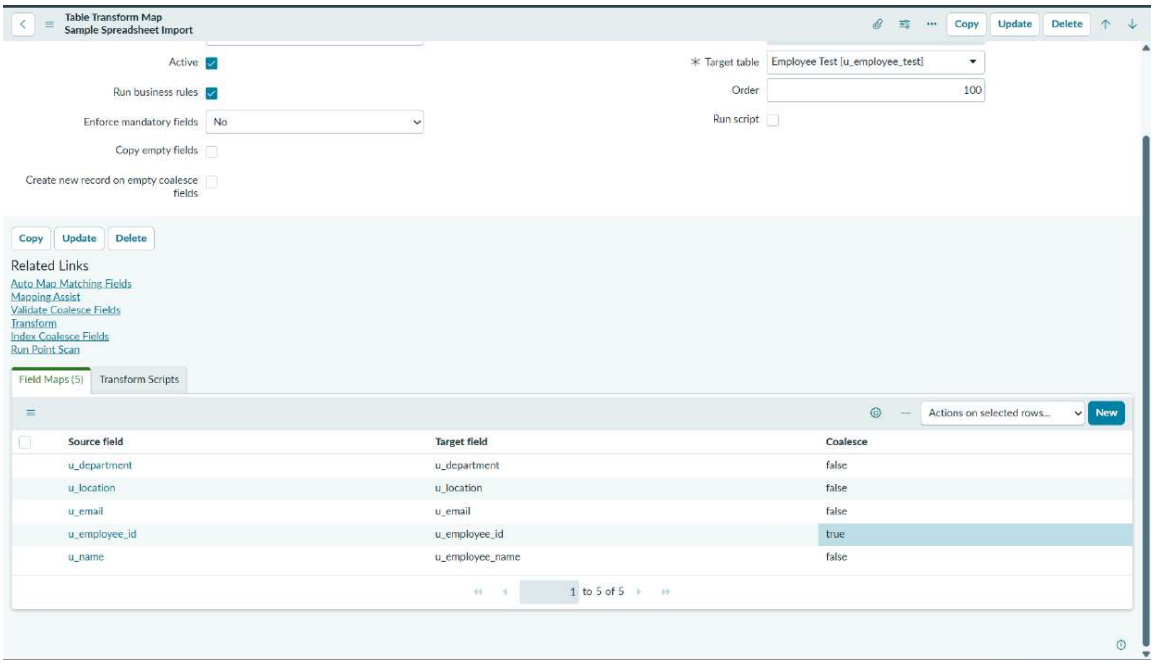
Source document: Phase 8 - Department, location and employee list reports.

Source document: Phase 9 - Employee Analytics Dashboards and sharing.

Uploaded Project Screenshots

Screenshots from Screenshot 2026-10-02 102233.pdf. The original explanatory diagram is retained.

Field mapping and Employee ID coalesce



Employee Analytics Dashboard

